

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-III Semester (2025-2026)
Faculty Name	Mohd. Razia Alangir Banu
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Roll Number	2520030485
Branch	CSE
Course Outcome	Apply Binary Search Trees (BST) and self-balancing AVL Trees to construct, insert, delete, and search hierarchical datasets, demonstrating correct rotation operations and $O(\log n)$ performance guarantees in real-world file indexing and dictionary scenarios.

Case Study Topic: EduGraph – Adaptive Learning & Course Recommendation Platform

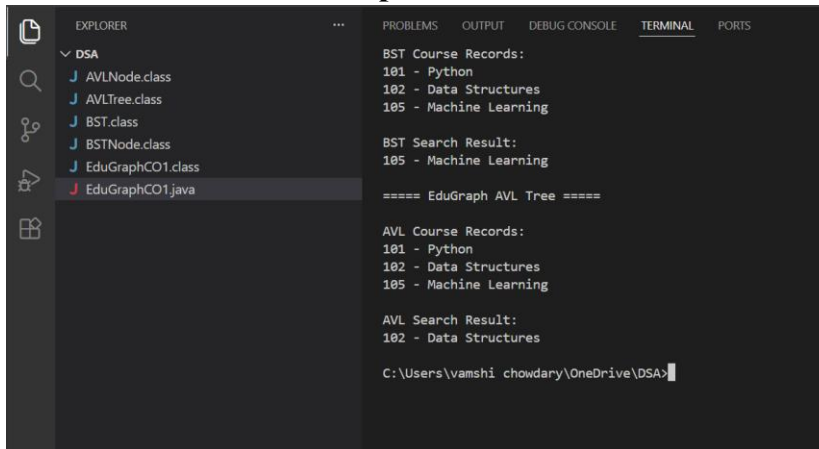
Work Done Summary:

Implemented BST and AVL Trees to manage student and course data efficiently in the EduGraph platform. AVL rotations were used to maintain balanced trees and ensure fast searching, insertion, and deletion with $O(\log n)$ performance. This improved course indexing and student record management.

Implementation Details:

1. Implemented BST for storing and searching student and course records.
2. Used AVL Trees to maintain balanced hierarchical data structures.
3. Performed insertion, deletion, and search operations efficiently.
4. Applied AVL rotations (Left, Right, Left-Right, Right-Left) to avoid imbalance.
5. Achieved $O(\log n)$ time complexity for faster data retrieval and indexing.

Result: Screenshots / Output



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BSA Course Records:
101 - Python
102 - Data Structures
105 - Machine Learning

BST Search Result:
105 - Machine Learning

===== EduGraph AVL Tree =====

AVL Course Records:
101 - Python
102 - Data Structures
105 - Machine Learning

AVL Search Result:
102 - Data Structures

C:\Users\vamshi chowdary\OneDrive\DSA>
```

The program successfully implemented BST and AVL Trees for managing course records in the EduGraph platform. AVL Tree maintained balanced structure using rotations and provided efficient searching with $O(\log n)$ performance.

GitHub Link:

<https://github.com/geethika04-dot/DSA-2>

NOTE: The code and Screenshots and PDF have been uploaded in the above link, please check it there.

Student Signature:

Faculty Signature: